



WORKING FROM HEIGHTS LEARNER GUIDE

Working at height – hazards and control

Introduction

Work at height accounts for about 50–60 deaths – more than any other workplace activity – and 4000 injuries each year. During the first two weeks of June 2003, inspectors from the Health and Safety Executive visited 1446 construction sites and stopped all work at one quarter of them due to concerns about the level of risks of falls from height. Another 5% of sites visited were issued with improvement notices. Other problems included a lack of or inadequate toe boards and intermediate guard rails on scaffolding and working platforms.

Attempts have been made to address these concerns by the introduction of the Work at Height Regulations which apply to all operations carried out at height not just construction work so that they are also relevant to, for example, window cleaning, tree surgery, maintenance work at height and the changing of street lamps.

The Work at Height Regulations affect approximately 3 m workers where working at height is essential to their work. Amongst other measures, the height of guard rails on scaffolds is increased by 40 mm to 950 mm thus recognizing that the average height of workers has increased over the last 40 years. The design of construction projects can also reduce health and safety hazards and accident rates. A study of accident data on falls from height over the five-year period from 1996 until 2001 emphasized the importance of building design in the elimination of work at height hazards. HSE recently published the following information:

In 2001/02, 254 painters and decorators were seriously injured due to falls from height, 89 of which were falls over 2 m

In 2002/03, 223 painters and decorators were seriously injured due to falls from height, 103 of which were falls under 2 m

In 2003/04, 245 painters and decorators were seriously injured due to falls from height, six of the falls were fatal.

The Work at Height Regulations

They have a minimum height 1.2m requirement for work at height. They include all work activities where there is a need to control a risk of falling a distance liable to cause personal injury. This is regardless of the work equipment being used, the duration of the work at height involved, or the height at which the work is performed. It includes access to and egress from a place of work. It would therefore include:

- working on a scaffold or from a mobile elevated work platform (MEWP)
- sheeting a lorry or dipping a road tanker
- working on the top of a container in docks or on a ship or storage area
- tree surgery and other forestry work at height using cradles or rope for access to a building or other structure like a ship under repair
- climbing permanent structures like a gantry or telephone pole
- working near an excavation area or cellar opening if a person could fall into it and be injured
- painting or pasting and erecting bill posters at height
- work on staging or trestles, e.g. for filming or events using a ladder/stepladder or kick stool for shelf filling, window cleaning.

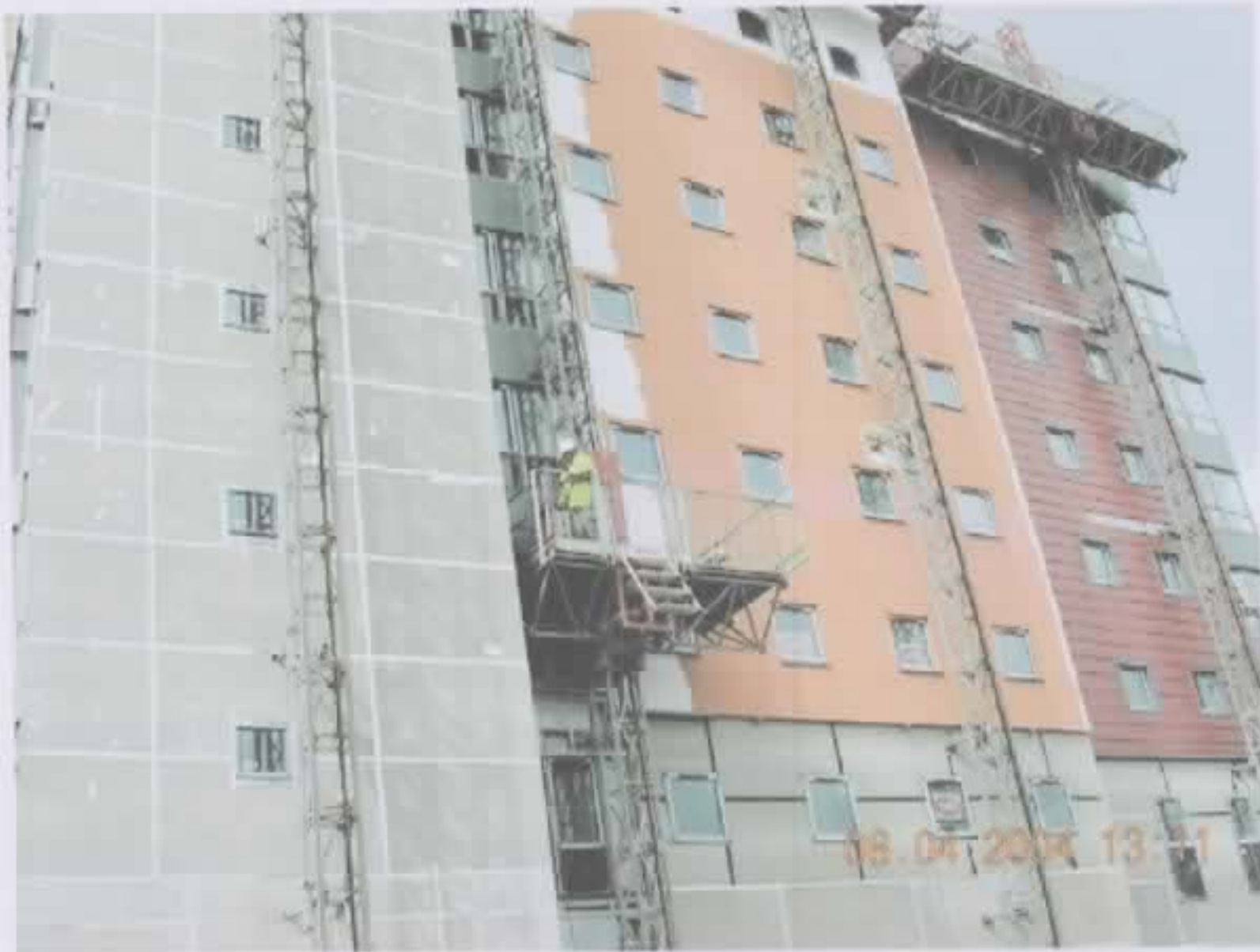


Figure 8.1 Working at height – mast climbing platforms.



Figure 8.2 Working platform, tower scaffolds and bridging units.

using manriding harnesses in ship repair, off-shore or steeplejack work
working in a mine shaft or chimney
work carried out at a private house by a person employed for the purpose, e.g. painter and decorator (but not if the private individual carries out work on their own home).

However it would not include:

- slips, trips and falls on the same level
- falls on permanent stairs if there is no structural or maintenance work being done
- work in the upper floor of a multi-storey building where there is no risk of falling (except separate activities like using a stepladder).

At the centre of the Regulations (Regulation 6), the employer is expected to apply a three-stage hierarchy to all work which is to be carried out at height. The three steps are the avoidance of work at height, the prevention of workers from falling and the mitigation of the effect on workers of falls should they occur.

The Regulations require that:

- work is not carried out at height when it is reasonably practicable to carry the work out safely other than at height (e.g. the assembly of components should be done at ground level)
- when work is carried out at height, the employer shall take suitable and sufficient measures to prevent, so far as is reasonably practicable, any person falling a distance liable to cause injury (e.g. the use of guard rails)
- the employer shall take suitable and sufficient measures to minimize the distance and consequences of a fall (collective measures, e.g. air bags or safety nets, must take precedence over individual measures, e.g. safety harnesses).

The risk assessment and action required to control risks from using a kick stool to collect books from a shelf should be simple (e.g. not overloading, not overstretching, etc). However, the action required for a complex construction project would involve significantly greater consideration and assessment of risk. A summary of the Regulations is given in Chapter 20.

Construction hazards and controls from working at height, Construction Reg 8

8.3.1 Safe place of work

Safe access to and egress from the place of work is essential when working at height. All working platforms,

ladders, scaffolds, gangways and passenger and materials hoists must be safe for use and be subject to regular inspection. Working platforms should also be kept as clear and clean as possible to avoid slip and trips and the accidental loss of material to the ground below.

8.3.2 Typical work activities and injuries associated with falls from height

Falls from height can result in fractures, serious head injuries and, in some cases, death. It is, therefore, essential that special care is taken to protect workers when they are working at height.

Common construction work activities which are done at height include brick or block laying, roofing, steel work erection, rendering, cladding, high pressure water-jetting, grit-blasting, concrete repairs, painting and some demolition work. For all these activities safe systems of work are essential. General hazards, such as dust, hazardous substances, noise, electricity, vibrations and noise can also be present due to the nature of the work being carried out at height.

One particular form of such activity which can be particularly hazardous is the use of false-work. False-work is a temporary structure used to support a non-self-supporting structure during its construction or refurbishment. An example is the wooden structure to support a brickwork arch which is being built. Only competent persons should plan, erect, load or dismantle false-work. The collapse of false-work structures is common and, when occurring at height, can cause serious injury. Most accidents caused by the collapse of false-work are caused by the lack of liaison between the various trades using it and its suppliers or erectors.

The cleaning of buildings and monuments involves the use of hazardous techniques such as grit-blasting or high pressure water jets, which is often done from scaffolding or even ladders. With both these techniques, care is needed to protect the workers, the occupants of the building and passers-by from hazards such as noise, dust, falling debris and possible flooding of walkways. The equipment must be maintained and inspected regularly and operatives provided with training, supervision and suitable personal protective equipment (goggles, ear defenders and gloves) and waterproof clothing. Other measures will be necessary, such as the boarding up of windows which may require the provision of temporary lighting inside the building. In the case of water-jetting, the scaffold should be enclosed in tarpaulin sheets and a channel system should be used to enable water to run off. Finally, stairwells and other holes in floors are another source of falls from height and measures must be taken to address these hazards. Such measures



Figure 8.3 Industrial roof work: use of safety nets.

include handrails on stairs, guard rails to stairwells and lift shafts, and ensuring that other holes in floors are similarly protected or covered. However, adequate levels of lighting and good housekeeping are also important as is a high standard of supervision and monitoring control.

8.3.3 Protection against falls

Falls from a height are the most common cause of serious injury or death in the construction industry. The Work at Height Regulations require that suitable and sufficient steps be taken to prevent any person falling a distance that would be liable to cause personal injury and specifies that the maximum unprotected gap between the toe and guard rail of a scaffold is 470 mm. This implies the use of an intermediate guard rail or infill. It also specifies requirements for personal suspension equipment and means of arresting falls (such as safety nets).

When working at height, a hierarchy of measures should then be followed, to prevent falls from occurring. These measures are:

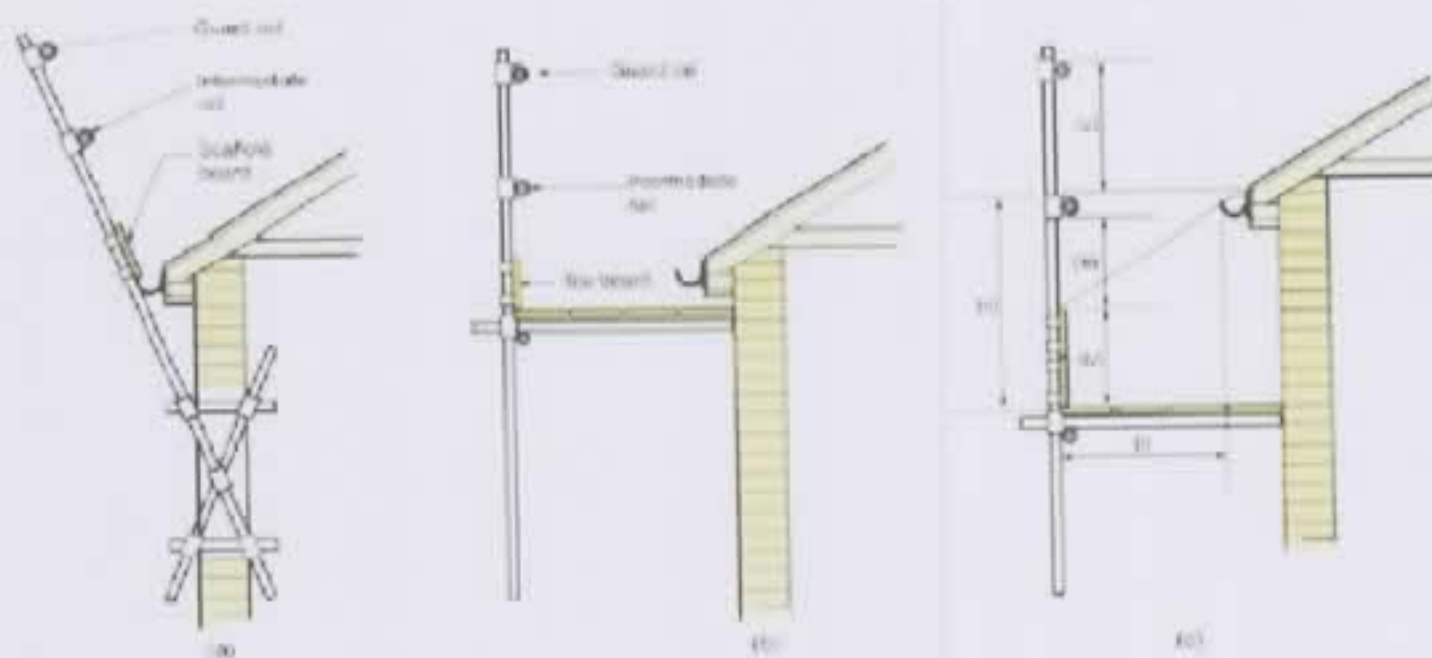
- avoid working at height, if possible the provision of a properly constructed working platform, complete with toe boards and guard rails if this is not practicable or where the work is of short duration, suspension equipment should be used and only when this is impracticable
- collective fall arrest equipment (air bags or safety nets) may be used where this is not practicable, individual fall restrainers (safety harnesses) should be used only when none of the above measures are practicable, should ladders or stepladders be considered.



Figure 8.4 Proper precautions must always be taken when working on or near fragile roofs.

8.3.4 Fragile roofs and surfaces

Work on or near fragile surfaces is also covered by the WAHR (see summary in Chapter 20). Roof work, particularly work on pitched roofs, is hazardous and requires a specific risk assessment and method statement (see Chapter 6 for a definition) prior to the commencement of work. Particular hazards are fragile roofing materials, including those materials which deteriorate and become more brittle with age and exposure to sunlight, exposed edges, unsafe access equipment and falls from girders, ridges or purlins. There must be suitable means of access such as scaffolding, ladders and crawling boards. Suitable barriers, guard rails or covers where people work near to fragile materials and roof lights. Suitable warning signs indicating that a roof is fragile, should be on display at ground level. There are other hazards associated with roof work – overhead services and obstructions, the presence of asbestos or other hazardous substances, the use of



Sloping roof edge protection: typical arrangement in conventional tube and fittings

- (a) Supported from window opening
- (b) Working platform below the eaves
- (c) Top lift of a scaffold. Dimensions should be as follows:
 - (i) Working platform minimum width 600 mm
 - (ii) Minimum 950 mm
 - (iii) Maximum gap 470 mm
 - (iv) To rise to the line of the roof slope with a minimum height of 150 mm
 - (v) Gap between rails no more than 470 mm

Figure 8.5 Typical sloping roof edge protection. Barriers shown in (a) can be useful where spaces is limited, but they are not capable of sustaining loads so large as (b) and (c) which also provide a working platform.

equipment such as gas cylinders and bitumen boilers and manual handling hazards.

It is essential that only trained and competent persons are allowed to work on roofs and that they wear footwear having a good grip. It is good practice to ensure that a person does not work alone on a roof.

8.3.5 Protection against falling objects

This topic is covered in Chapter 7 because it is a general site issue. However, it is particularly relevant when working at height and is also now covered by WAHR (see summary in Chapter 20). A study by the HSE has indicated that over a five year period, 44 construction workers and members of the public (including children) were killed by insecure loads, unsecured equipment and pieces of plant falling on them. Over the same period 59 were fatally injured by the collapse of structures or parts of structures. Whilst 37 of these deaths were caused by accidents during demolition work, the remainder occurred during routine construction work. As mentioned, workers and members of the public need to be protected from the hazards associated with falling objects by the use of covered walkways or suitable netting to catch falling debris. Waste material should be brought to ground level by the use of chutes or hoists and not thrown from height. Minimal quantities of

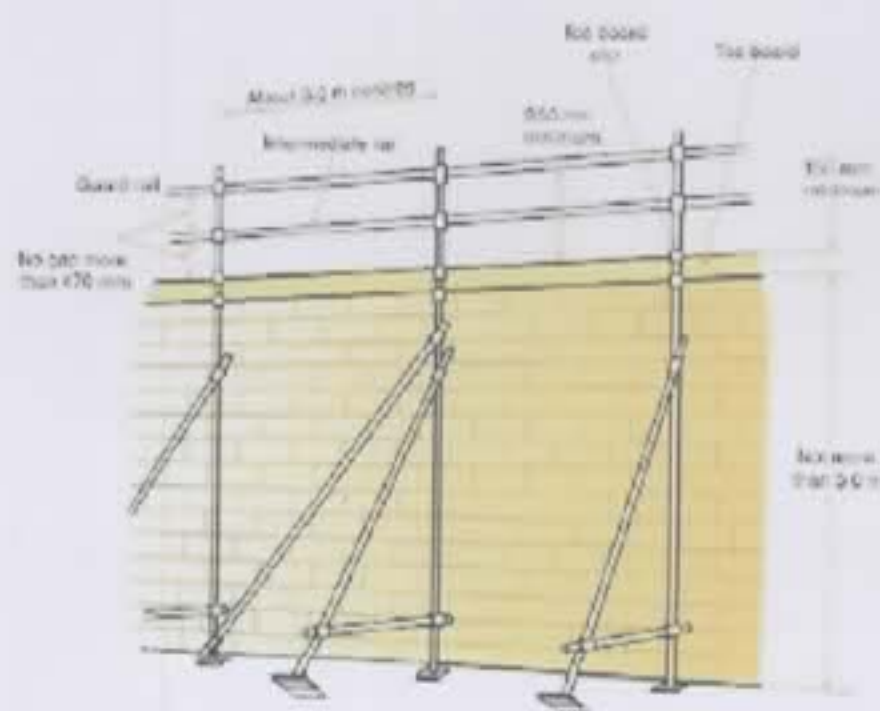


Figure 8.6 Flat roof edge protection supported at ground level. Ground level support allows work up to the roof edge without obstruction.

building materials should be stored on working platforms.

Head protection is mandatory on site for workers and visitors. Several of the lives lost due to accidents from falling materials could have been saved if hard hats had been worn.

8.3.6 The effect of deteriorating materials and weather

As many materials age, they become brittle, weaker and less stable. This has already been mentioned as a hazard associated with fragile roofs. It is also a problem in other situations found in construction work, such as refurbishment work and the various cladding materials (including asbestos) used inside and outside buildings. The well publicized concrete 'cancer', where concrete degenerates to a powder leading to serious structural weaknesses, has occurred in several relatively young reinforced concrete structures and causing either very expensive repairs or total rebuilds. Whilst all materials will deteriorate to some extent with age, good design, maintenance, construction methods (using the correct mix ratios for mortars and concrete) and type of cement should significantly reduce the rate of deterioration. Weather is also a significant factor which can increase the hazards associated with working at height. High winds can affect the stability of scaffolding and ladders as has been mentioned elsewhere in this chapter. They can affect the work process itself by displacing tools and materials and cause injuries to persons working at height or those below the workings. Such winds, if strong enough, can also lift roofing tiles or sheets particularly if the roof is only partially completed. Such problems often occur outside normal working hours causing injuries to members of the public (including trespassers) and damage to neighbouring properties. It is very important to ensure that the construction work is protected after working hours especially if bad weather is forecast. Lightning is another weather hazard and it is inadvisable for work to continue on roofs or scaffolds when thunderstorms are approaching. The effects of lightning are discussed in more detail in Chapter 14. Even apparently benign weather can present health and safety hazards. Skin cancer rates due to excessive exposure to sunlight has increased significantly over the last two decades. Even if skin cancer does not occur, serious damage to the skin will result from regular exposure to strong sun over several years. Typical damage is wrinkled skin due to the breakdown of skin elasticity and the formation of red patches of skin. There are also concerns with the depletion of the ozone layer, that strong sunlight may cause serious deterioration in eye health, leading to possible problems such as arc eye and long term eye damage. Precautions such as the use of protective creams, sunglasses, and reduced exposure by wearing suitable clothing, are all essential controls. This topic is covered in more detail in Chapter 17. Finally, cold weather can also present hazards when working at height particularly when the cold weather is accompanied by wind which will produce a 'chill factor' and an apparent temperature that is lower than the actual

one. Guard rails will become colder and the reactions of workers much slower. Of course, cold weather will have an effect on construction processes, such as the mixing of cement and the laying of bricks. Workers should wear appropriate clothing and if the temperature is too low, work at height should cease.

8.3.7 Working over water

Where construction work takes place over water, steps should be taken to prevent people falling into the water and rescue equipment should be available at all times. If the work takes place adjacent to a river or canal, special precautions are needed for persons using a tow path. The Construction (Health, Safety and Welfare) Regulations 1996 cover any work where there is a risk to operatives of drowning. Working on or adjacent to water or when people have to pass near or across water on their way to work gives rise to a risk of drowning. People can also drown in other liquids such as slurries, chemicals and some free flowing foodstuffs in vats and silos.

At the planning stage of any work over or beside water, the following points need to be considered:

- ☐ the provision of suitable fencing to ensure that persons do not fall into the water by the provision of barriers or fences with toe boards
- ☐ the provision of life jackets, buoyancy aids and grab lines (fast flowing water like rivers)
- ☐ the provision of rescue equipment, lifebelts, harnesses, safety lines, and in some cases, a rescue boat
- ☐ where transport by water is required, arrangements must be made for safe landing stages and life jackets
- ☐ when boats are used they should be of suitable construction; under the control of a competent person; not overloaded and properly maintained
- ☐ the provision of rescue teams and procedures specific training for boat operatives, rescue teams and supervisors
- ☐ all precautions must also protect the safety of the public especially children
suitable security measures will be necessary to prevent the theft of rescue equipment and boats outside working hours
- ☐ weather, tides and flooding conditions must be taken into account when planning rescue measures and using boats as transport
- ☐ all openings, drains and vats must be covered, where possible.

Site supervision is required to ensure that all barriers, fencing and rescue equipment is provided before any work commences which could place personnel at risk

from drowning. The site supervisor will ensure that only authorized personnel alter barriers and operate rescue equipment and boats. The supervisor will also ensure that all rescue equipment is checked regularly and that any defective equipment is repaired or replaced immediately. He will ensure that personnel appointed as rescue teams are available during working periods and that replacement personnel are available when necessary.

There are several general hazards associated with working at height over or adjacent to water which are similar to those for any work done at height. These include falling from a height, scaffold stability, falling material, pedestrian safety, and possible collision with the scaffolding by boats. Many of these issues can be addressed by the provision of a properly designed scaffold, erected by competent persons and securely tied and provided with guard rails, toe boards and ladder access to working platforms. Other precautions include fall arrest equipment such as safety nets or harnesses, the non-overcrowding of the scaffold, the provision of rescue equipment such as lifebelts and the availability of a rescue boat under the control of a competent person, should someone fall into the water. Harnesses must be worn by those working on the scaffold in addition to the use of appropriate personal protective equipment. There should be a current register of workers and nobody should work alone on the scaffold.

For pedestrians, using perhaps a towpath, a protected access or safe diversion route must be provided. Signs, lighting and protective barriers will also be required to minimize the risk of boats colliding with the scaffold.

Finally, consideration must be given to health hazards which are associated with working near water. These include biological hazards such as leptospirosis and tetanus and general hazards to health associated with working on any construction site such as noise, vibration and musculoskeletal problems, electricity, mechanical hazards, flying particles, dust and contact with cement.

Appropriate control measures include disinfection, good standards of personal hygiene, the use of appropriate personal protective equipment, adequate washing and first-aid facilities and the provision of information, instruction, training and supervision.

Working above ground or where there is a risk of falling

8.4.1 Hazards and controls associated with working at height

Work at any height involves a risk of falling. The significance of injuries resulting from falls from height,

such as fatalities and other serious major injuries, have been dealt with earlier in this chapter as has the importance and legal requirements for head protection. Also covered were the many hazards involved in working at height including fragile roofs/surfaces and the deterioration of materials, unprotected edges and falling materials. Additional hazards include the weather and unstable or poorly maintained access equipment, such as ladders and various types of scaffold. Scaffolding must only be erected or modified by people who are trained and competent to do so. It should be constructed in accordance with recognized standards from components which are of adequate design and strength, and which are inspected at regular intervals. The scaffolding itself should be inspected by a competent person regularly.

The principal means of preventing falls of people or materials includes the use of fencing, guard rails, toe boards, working platforms, access boards, ladder hoops, safety nets and safety harnesses. Safety harnesses arrest the fall by restricting the fall to a given distance due to the fixing of the harness to a point on an adjacent rigid structure. They should only be used when all other possibilities are not practical.

8.4.2 Access equipment

There are many different types of access equipment but only the following four categories will be considered here:

- ladders
- fixed scaffolds
- mobile scaffold towers
- mobile elevated work platforms.



Figure 8.7 Ladders should be correctly angled one out for every four up.

Ladders

The Work at Height Regulations place a duty on employers to ensure that a ladder is used for work at height only if a risk assessment has demonstrated that the use of more suitable work equipment is not justified because of the low risk and:

- ☐ the short duration of use or
- ☐ existing features on the site which he cannot alter.

Falls from ladders caused 13 deaths in 2005 and a third of all major injuries caused by falls at work were due to falls from ladders (1200). Ten ladder accidents are reported to the HSE every day.

The main cause of accidents involving ladders is ladder movement whilst in use. This occurs when they have not been secured to a fixed point particularly at the foot. Other causes include over-reaching by the worker, slipping on a rung, ladder defects and, in the case of metal ladders, contact with electricity. The main category of ladder accidents is falls.

There are two common materials used in the construction of ladders – aluminium and timber. Aluminium ladders have the advantage of being light but should not be used in high winds or near live electricity. Timber ladders need regular inspection for damage and should not be painted since this could hide cracks and other defects, such as knots.

The following factors should be considered when using ladders:

- ensure that the use of a ladder is the safest means of access given the work to be done and the height to be climbed
- the location itself needs to be checked. The supporting wall and supporting ground surface should be dry and slip free. Extra care will be needed if the area is busy with pedestrians or vehicles
- the ladder needs to be stable in use. This means that the inclination should be as near the optimum as possible (1 in 4 ratio of distance from the wall to distance up the wall). The foot of the ladder should be tied to a rigid support. Weather conditions must be suitable (no high winds or heavy rain). The proximity of live electricity should also be checked. (This last point is important when ladders are to be carried beneath power lines)
- the ladder should be fitted with non-slip feet
- there should be at least 1 m of ladder above the stepping off point
- the work activity must be considered in some detail. Over-reaching must be eliminated and consideration given to the storage of paints or tools which are to be used from the ladder and any loads to be carried

up the ladder. The ladder must be matched to work required

- workers who are to use ladders must be trained in the correct method of use and selection. Such training should include the use of both hands during climbing, clean non-slippery footwear, clean rungs and an undamaged ladder
- ladders should be inspected (particularly for damaged or missing rungs) and maintained on a regular basis and they should only be repaired by competent persons
- the transportation and storage of ladders is important since much damage can occur at these times. They need to be handled carefully and stored in a dry place
- when a ladder is left secured to a structure during non-working hours, a plank should be tied to the rungs to prevent unauthorized access to the structure.

There are a variety of products available to ensure that ladders are stable during use by the provision of anchors for the ladder base or fixing mechanisms to the building. However, certain work should not be attempted using ladders particularly when the user cannot maintain a safe hand hold while carrying a load. This includes work where:

- a secure hand hold is not available
- the work is at an excessive height
- where the ladder cannot be secured or made stable
- the work is of long duration
- the work area is very large
- the equipment or materials to be used are heavy or bulky
- the weather conditions are adverse
- there is no protection from vehicles.

A particular type of timber ladder is the wooden pole ladder, which is used by many scaffolders. Pole ladders should be suitable for the job by being of the appropriate length and having complete integrity of the steel reinforcement. Prior to use, the ladder should be inspected as outlined earlier in this section and any damage to stiles, rungs or steel reinforcement must be identified as should any unauthorized repairs.

There have been several rumours that the Work at Height Regulations have banned the use of ladders. This is not true. Ladders may be used for access and it is legal to work from ladders. Ladders may be used when a risk assessment shows that the risk of injury is low and the task is of short duration or there are unalterable features of the work site and that it is not reasonably practicable to use potentially safer alternative means of access. Provided that the ladder is checked before each use,

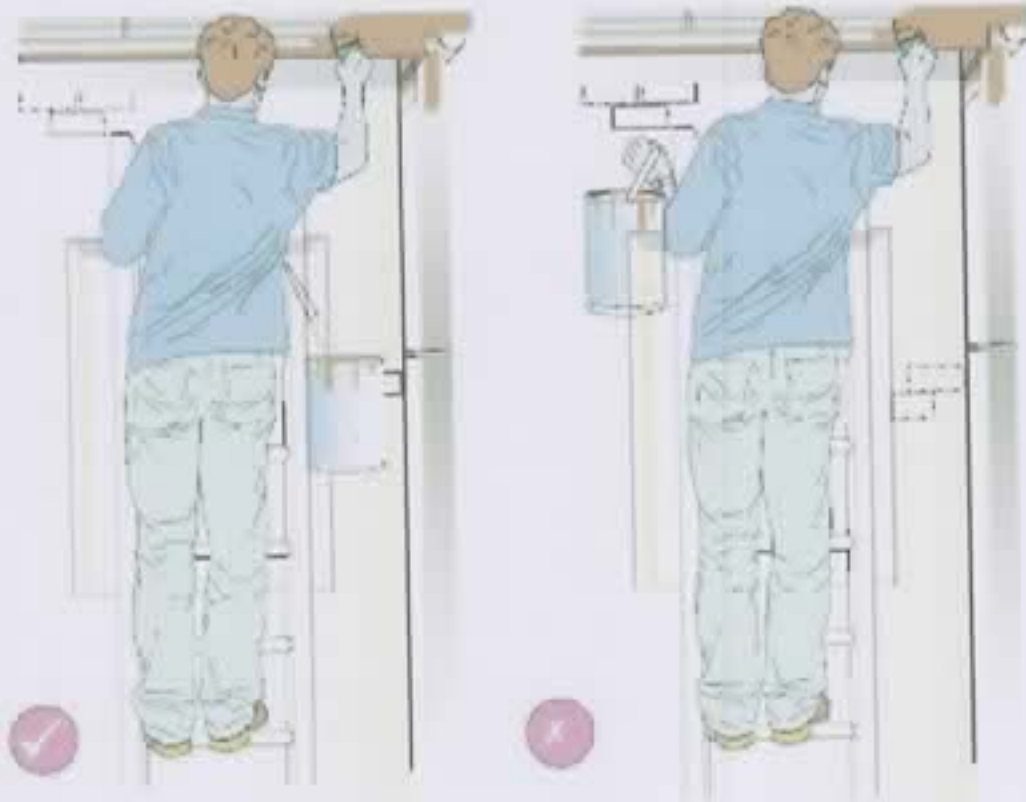


Figure 8.8a Attach paint cans and the like to the ladder

- Glass fibre BSEN131: 1993 Kite marked Industrial
- Step stools BS7377: 1994.



Figure 8.8b Pole ladder in use.

the work is of short duration, only light loads are to be carried, the ladder is stable and secured and users are trained and competent, then ladders may be used. More information on ladders and their use within the requirements of the Work at Height Regulations is available from the British Ladder Manufacturers Association. Ladders for industrial work in the UK should be marked to:

- Timber BS1129: Kite marked Class 1 Industrial
- Aluminium BS2037: 1994 Kite marked Class 1 Industrial

Stepladders, trestles and staging

Many of the points above for ladders apply to stepladders and trestles where stability and over-reaching are the main hazards. Neither of these should be used as a workplace above about 2 m in height unless proper edge protection is provided.

All equipment must be checked by the supervisor before use to ensure that there are no defects and must be checked at least weekly whilst in use on site. If a defect is noted, or the equipment is damaged, it must be taken out of use immediately. Any repairs must only be carried out by competent persons.

Supervisors must also check that the equipment is being used correctly and not being used where a safer method should be provided.

Where staging, such as a 'Youngmans' staging platform, is being used in roof areas, supervisors must ensure that only experienced operatives are permitted to carry out this work and that all necessary safety harnesses and anchorage points are provided and used. The main hazards associated with stepladders, trestles and staging are:

- unsuitable base (uneven or loose materials)
- unsafe and incorrect use of equipment (e.g. the use of staging for barrow ramps)
- overloading
- use of equipment where a safer method should be provided
- overhang of boards or staging at supports ('trap ends')

- use of defective equipment.

Stepladders and trestles must be:

- manufactured to a recognized industrial specification
- stored and handled with care to prevent damage and deterioration
- subject to a programme of regular inspection (there should be a marking, coding or tagging system to confirm that the inspection has taken place)
- checked by the user before use
- taken out of use if damaged – and destroyed or repaired
- used on a secure surface, and with due regard to ensuring stability at all times
- kept away from overhead cables and similar hazards.

The small platform fitted at the top of many stepladders is designed to support tools, paint pots and other working materials. It should not be used as a working place unless the stepladder has been constructed with a suitable handhold above the platform. Stepladders must not be used if they are too short for the work being undertaken, or if there is not enough space to open them out fully.

Platforms based on trestles should be fully boarded, adequately supported (at least one support for each 1.5 m of board for standard scaffold boards) and provided with edge protection when the platform is 2 m or higher.

Fixed scaffolds

It is quicker and easier to use a ladder as a means of access but it is not always the safest. Jobs such as painting, gutter repair, demolition work or window replacement should normally be done using a scaffold. Scaffolds must be capable of supporting building workers, equipment, materials, tools and any accumulated waste. A common cause of scaffold collapse is the 'borrowing' of boards and tubes from the scaffold thus weakening it. Falls from scaffolds are often caused by badly constructed working platforms, inadequate guard rails or climbing up the outside of a scaffold. Falls also occur during the assembly or dismantling process.

There are two basic types of external scaffold:

- independent tied – these are scaffolding structures which are independent of the building but tied to it often using a window or window recess. This is the most common form of scaffolding; and
- putlog – this form of scaffolding is usually used during the construction of a building. A putlog is a scaffold tube which spans horizontally from the scaffold into the building – the end of the tube is

flattened and is usually positioned between two brick courses.

The important components of a scaffold have been defined in a guidance note issued by the HSE as follows.

Standard: an upright tube or pole used as a vertical support in a scaffold.

Ledger: a tube spanning horizontally and tying standards longitudinally.

Transom: a tube spanning across ledgers to tie a scaffold transversely. It may also support a working platform.

Bracing: tubes which span diagonally to strengthen and prevent movement of the scaffold.

Putlog: a tube which is flattened at its end and spans from a ledger to the wall of a building. A scaffold secured to a building in this way is known as a putlog scaffold.

Guard rail: a horizontal tube fitted to standards along working platforms to prevent persons from falling.

Toe boards: these are fitted at the base of working platforms to prevent persons, materials or tools falling from the scaffold.

Raker: an angled or inclined load bearing tube used to support a cantilevered scaffold working platform. It is fixed to the edge of the working platform at one end and the wall of the building at the other.

Base plate: a square steel plate fitted to the bottom of a standard at ground level.

Sole board: normally a timber plank positioned beneath at least two base plates to provide a more uniform distribution of the scaffold load over the ground.

Fans: specially designed platforms which prevent debris and other materials falling on people passing or working below the scaffold.

The following factors must be addressed if a scaffold is to be considered for use for construction purposes:

- scaffolding must only be erected and dismantled by competent people who have attended recognized training courses. Any work carried out on the scaffold must be supervised by a competent

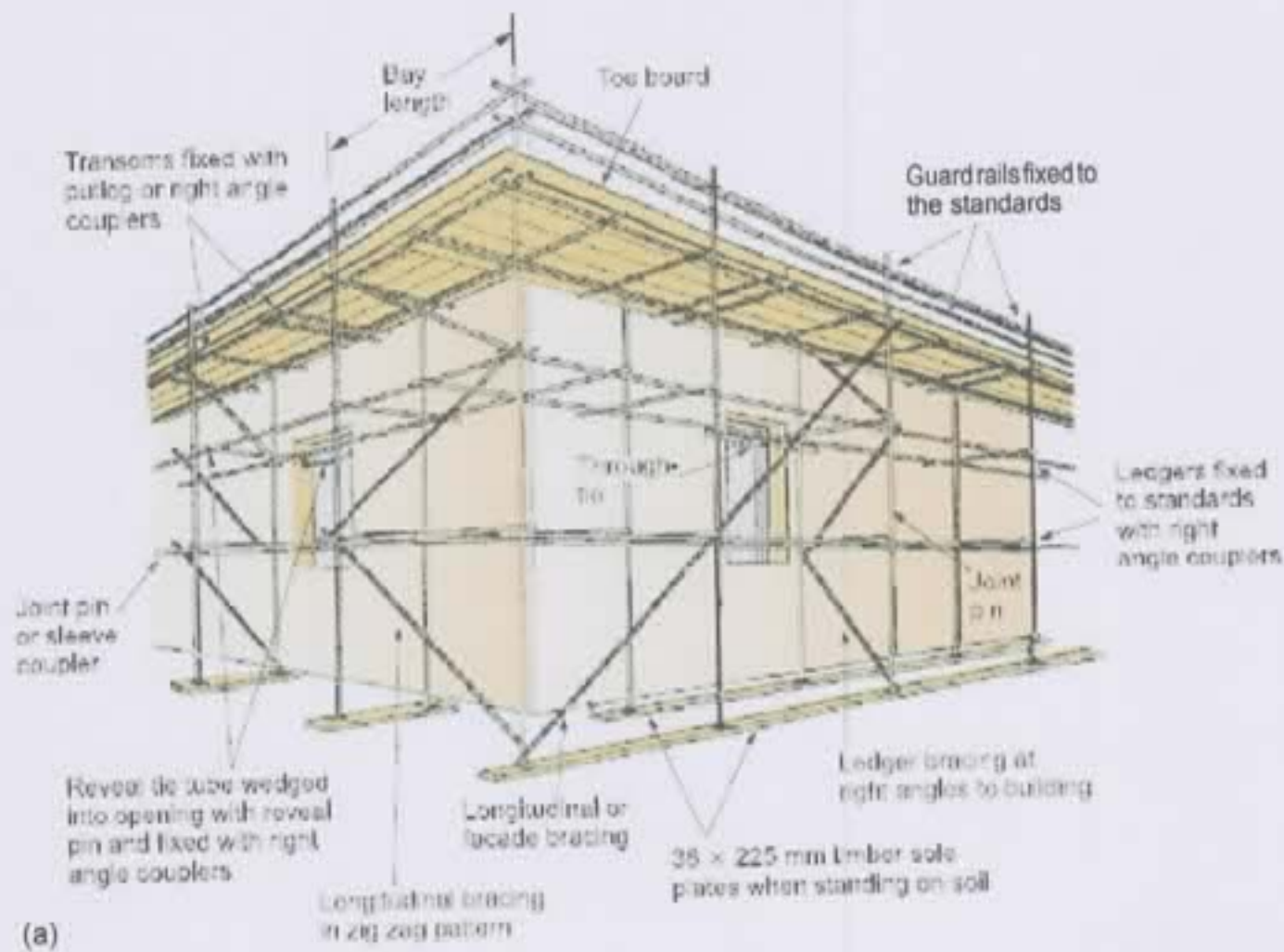


Figure 8.9 (a) Typical independent tied scaffold; (b) fixed scaffold left in place to fit the gutters.

- person. For complex scaffolds, a written plan may be required. Any changes to the scaffold must be done by a competent person
- in busy town centres, the erection and dismantling of the scaffold should take place during quiet times
- the scaffold must be designed to carry all the required loads and use only sound fittings and materials
- adequate toe boards, guard rails and intermediate rails must be fitted to prevent people or materials from falling

- appropriate bay lengths and widths, bracing and tie-in arrangements together with edge and stop-end protection
- the scaffold must rest on a stable surface, uprights should have base plates and timber sole boards if necessary
- the scaffold must have safe access and egress
- work platforms should be fully boarded with no tipping or tripping hazards
- the scaffold must be maintained, cleaned and have proper arrangements for waste disposal
- the scaffold should be sited away from or protected from traffic routes so that it is not damaged by vehicles
- if the scaffold must be erected near to vehicle roadways then barriers or fenders should be fitted around its base
- the scaffold should be properly braced and secured to the building or structure
- overloading of the scaffold must be avoided
- the public must be protected at all stages of the work and the appropriate highway authority contacted before a scaffold is erected beside a public thoroughfare
- regular inspections of the scaffold must be made and recorded.

There have been many examples of scaffold collapse causing serious injuries to workers and members of the public. Factors which may affect the stability of an independently tied scaffold include:

- poor, incompetent erection and/or a lack of regular inspections
- the strength of the supporting ground or foundation
- the proximity to the scaffold of any excavation work
- the effect of excessive surface water in weakening the scaffold foundation
- unauthorized alteration by incompetent persons (perhaps by vandals)
- the use of incorrect or damaged fittings during the erection or extension of the scaffold
- the overloading of the scaffold
- adverse weather conditions
- the sheeting in of the scaffold without the use of extra ties
- vehicular impact.

With all scaffolds, brick guards and debris netting should be used to prevent materials and equipment falling to the ground.

For putlog scaffolds, all the previously stated safety requirements apply and some additional ones. The

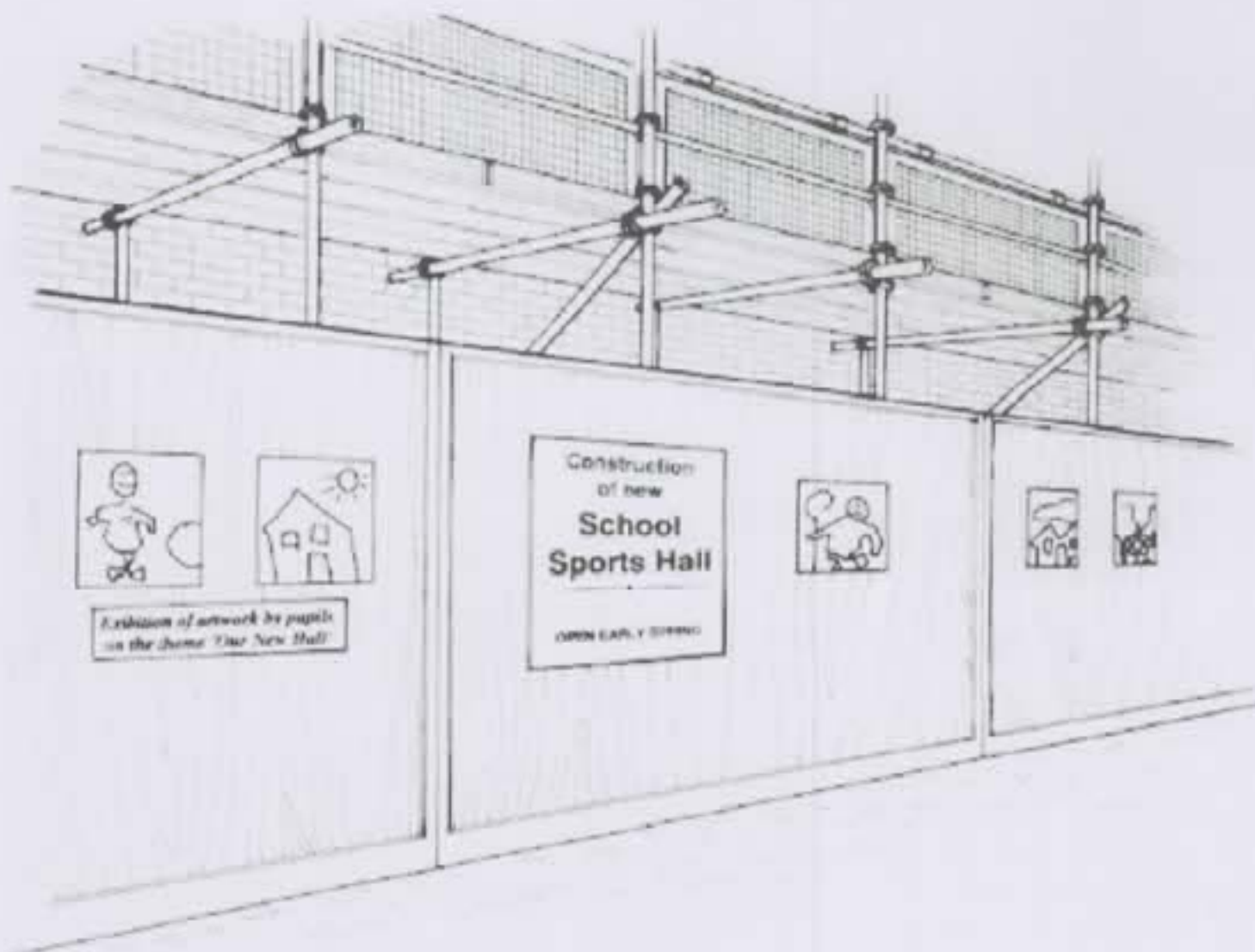


Figure 8.10 Fan scaffold. Panels are fixed around the base of the scaffolding to prevent children from climbing.

putlog ends, which must be properly flattened, should be fully inserted and securely bedded into well constructed brickwork. It is recommended that a putlog scaffold should not exceed 45 m in height.

When the work is to take place in a busy area,

all the normal safety features of scaffolding apply, but, in addition, special care should be taken with ladder access, debris netting and fans. Appropriate traffic signs (and possibly traffic lights), cones, barriers and lighting arrangements at night will also be necessary.

All scaffolds should have an assembly, use and dismantling plan. Before accepting a scaffold erected by a specialist scaffolding sub-contractor for use, the site supervisor must check the scaffold and obtain a handing-over certificate.

The site supervisor will ensure that all scaffolds are erected in accordance with the standards outlined above and, at the beginning of each week, he should inspect the scaffold and ensure that any defect is rectified. A report of the inspection and action taken should be entered in the site inspection register. A similar inspection should also be carried out after high winds or other adverse weather conditions.

All materials used for scaffolding must be provided in accordance with the relevant British Standards and checked before use by a scaffolder. Such materials must be properly stored and maintained on the site.

Any scaffold that is being erected, altered or dismantled, or otherwise not suitable for use by employees, must have a warning notice erected stating that it must not be used.

All scaffolds must be checked at the end of each working day to ensure that access to the scaffold by children has been prevented.

Requirements for scaffold erectors

All scaffolds must only be erected by scaffolders who are competent and experienced to undertake the work. All the requirements, which were outlined in Chapters 3 and 7, apply equally to scaffolders. Only scaffolding firms who have a good reputation and accident record gained from earlier contracts should be employed to erect scaffolds. Scaffolders should be supervised throughout the erection of a scaffold and they should be capable of inspecting scaffolding materials and equipment before it is used. They should be equipped with personal lifelines which must be attached at all times while working at height. All scaffolders must have attended a specialist scaffolding course and be able to provide documentary evidence that they have successfully completed the course.

Design of loading platforms

A loading platform is a particular working platform on a scaffold onto which materials and equipment is loaded to

further the construction project. A working platform is part of the scaffold or similar structure on which people work, walk and stand. It must be designed so that it can carry workers, their tools and working materials safely. The platform is built by laying scaffold boards across the transoms and, normally, at least three support points are required for each board. There must be a slight overhang (50 to 150 mm) across a support at each end of the board. Boards are normally butt jointed. Toe boards and guard rails, at least 950 mm above the working platform, must be fitted inside the standards along the edge of the working platform. Ladders used to access the loading platform must extend at least 1 m above the level of the platform.

All stacking of materials and equipment, such as bricks, must be as close to standards as possible to enable the added weight to be more evenly distributed throughout the scaffold. The boards will need some additional fixing to the structure so that they are protected from lifting during high winds.

The width of the platform will vary with the type of work being undertaken from a minimum of three boards to a maximum of seven (when trestles are to be used on the platform).

The following points are relevant to the design of all working platforms whether loading or not:

- they must be wide enough (at least 600 mm) to allow people to pass each other and for any required equipment or materials to be carried and stored
- they must be free of openings and trip hazards
- if members of the public are able to pass beneath, a double boarded platform with a sheet of polythene placed between the boards may be necessary to prevent debris or tools falling on them
- the principal hazards associated with working platforms are overloading, the fall of materials, tools and equipment and slips, trips and falls of people working on the platform
- good housekeeping and supervision are essential on all working platforms.

Finally, if it is likely that heavy loads are to be stored on a working platform, the scaffolding firm should be informed so that special support may be designed into the scaffold structure.

Scaffold hoists

Hoists are widely used to transport materials to higher working levels. It is important that hoists are well maintained, inspected regularly and in a good working order. They should be positioned on firm and stable ground, erected following the manufacturer's instructions and properly protected at either end. The hoist should be marked with its safe working load and its control functions.

The following applies to the use of all hoists whether they are mobile or permanent. The main hazards associated with hoist operations are:

- overloading of the hoist
- overloading or incorrect use of lifting gear
- automatic safe load indicator not working
- failure of equipment due to lack of maintenance
- incorrect positioning of lifting appliance
- insecure attachment of load and falls from the platform
- unstable slinging of loads, hook not over centre of gravity
- contact with overhead electricity cables
- lack of operator training
- no banksman used when drivers view is obscured
- incorrect signals.

Only hoists which are suitable for the site and capable of lifting the required loads should be selected and all personnel working with or near lifting appliances must wear hard hats. All hoists must be secured and left in a safe condition at the end of each working period so that any child trespassers are protected.



Figure 8.11 Hoist with interlocked gates.

The following additional precautions should also normally be taken:

- the controls should be set so that the hoist can be operated from one position only (normally ground level) and that all the landing levels are clearly visible by the operator from the operating position thus preventing people from being struck by the platform or other moving parts
- the hoistway must be enclosed at places where people might be struck, for example, working platforms or window openings
- gates must be provided at all landings and at ground level
- the hoistway must be fenced wherever people could fall down it
- the gates at landings are kept closed except during loading and unloading and they should be secure and not free to swing into the hoistway
- the edges of the hoist platform and the landing should be as close together as possible so that there is no significant gap between for debris to fall through
- loads, such as wheelbarrows, should be securely chocked and not overfilled
- loose loads, such as bricks, should be carried in proper containers or a hoist with an enclosed platform should be used
- the platform must not be overloaded and should be clearly marked with its safe working load
- the hoist must be erected by trained and experienced people who will follow the manufacturer's instructions and ensure that it is properly secured to the supporting structure
- the hoist operator must be properly trained and be competent
- loads must be evenly distributed on the hoist platform
- the hoist must be thoroughly examined and tested after erection, substantial alteration or repair and at six-month intervals
- regular inspections should be carried out at least on a weekly basis and the results recorded.

Finally, no one should be allowed to ride on a goods hoist and a notice, which states this prohibition, should always be posted.

Prefabricated mobile scaffold towers

Mobile scaffold towers are frequently used throughout industry. It is essential that the workers are trained in their use since recent research has revealed that 75% of lightweight mobile tower scaffolding is erected, used, moved or dismantled in an unsafe manner.

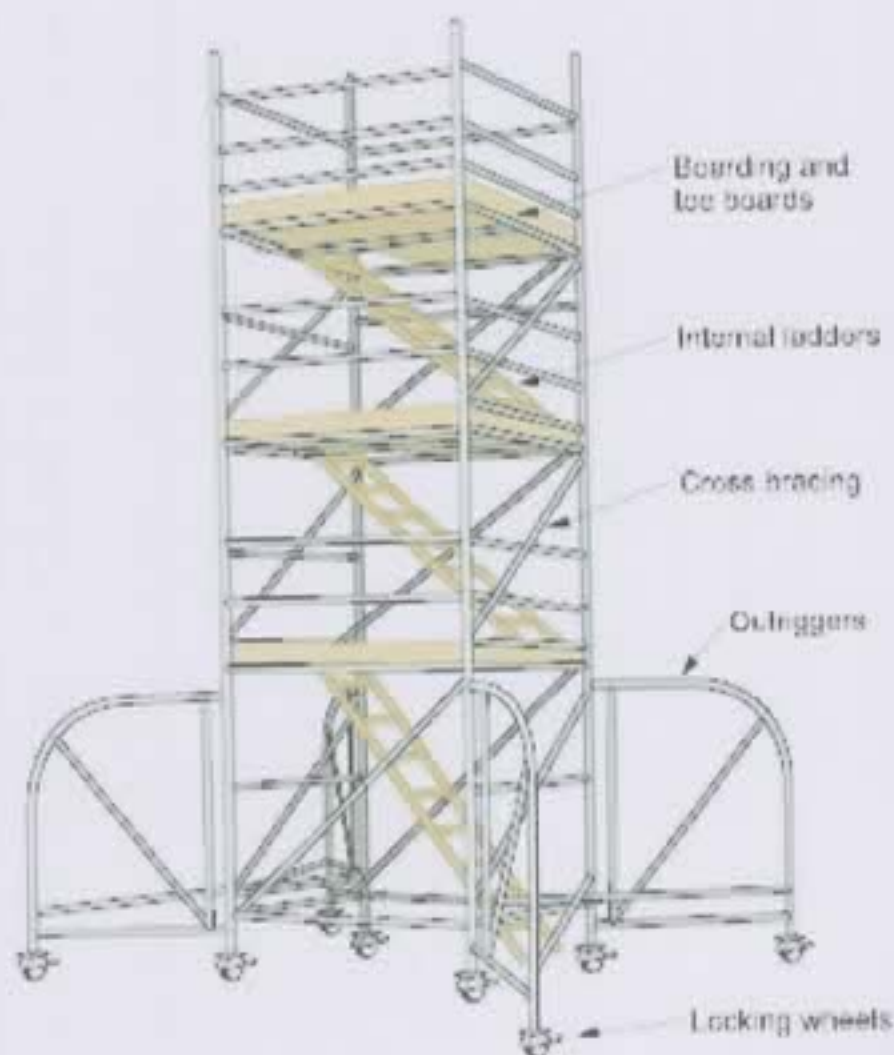


Figure 8.12 Typical prefabricated tower scaffold.

The following points must be considered when mobile scaffold towers are to be used:

- towers should only be used on level, firm and stable ground
- the selection, erection, dismantling and inspection of mobile scaffold towers must be undertaken by competent and trained persons with maximum height to base ratios not being exceeded
- no person must be permitted to erect, alter or dismantle any mobile tower scaffold unless authorized by the site supervisor
- diagonal bracing and stabilizers should always be used
- access ladders must be fitted to (not leaning on) the narrowest side of the tower or inside the tower and persons should not climb up the frame of the tower
- all wheels must be locked whilst work is in progress and all persons must vacate the tower before it is moved
- all operatives, who are required to use mobile tower scaffolds, will be instructed in their safe use and movement
- outriggers or stabilizers must be extended where applicable
- tower scaffolds must not be used or moved on sloping, uneven or obstructed surfaces
- tower scaffolds must only be used in the vertical position
- the tower working platform must be fully boarded, fitted with guard rails and toe boards and not overloaded
- towers must be tied to a rigid structure if exposed to windy weather or to be used for work such as jet blasting
- persons working from a tower must not over-reach or use ladders from the work platform
- safe distances must be maintained between the tower and overhead power lines both during working operations and when the tower is moved
- the tower should be inspected on a regular basis and a report made.

There are two approved methods for the erection of a prefabricated tower scaffold – an advance guard rail system and the ‘through the trap’ (3T) system. These systems have been developed by HSE and the Prefabricated Access Suppliers’ and Manufacturers’ Association and are described in detail on the HSE information sheet ‘Tower scaffolds’ CIS No 10 (Rev 4). The main reasons that a mobile tower scaffold may become unstable are:

- the brakes are defective
- it was not erected by a competent person or unauthorized adjustments were made
 - ☐ it was erected on ground that was neither firm nor level
 - ☐ the height of the tower exceeded the recommended base to height ratio
 - ☐ the failure to fit and/or use outriggers when they were required
 - ☐ the safe working load was exceeded
 - ☐ the movement of the tower either with persons and/or materials on the working platform or under windy conditions
 - ☐ collision with vehicles or other plant.

The recommended height to base ratio varies from 3:1 to 3.5:1, with a base not less than 1 m, dependent on whether the tower is to be used outside or inside the building. The ratio is lower for outside use due to possible wind effects. The ratios quoted should be treated as a guide and the manufacturer’s instruction should always be consulted. Mobile scaffold towers should be inspected regularly as outlined later in this chapter.

Mobile elevated work platforms (MEWPs)

Mobile elevated work platforms are very suitable for many different types of high level work such as changing light bulbs in a warehouse. Examples of such

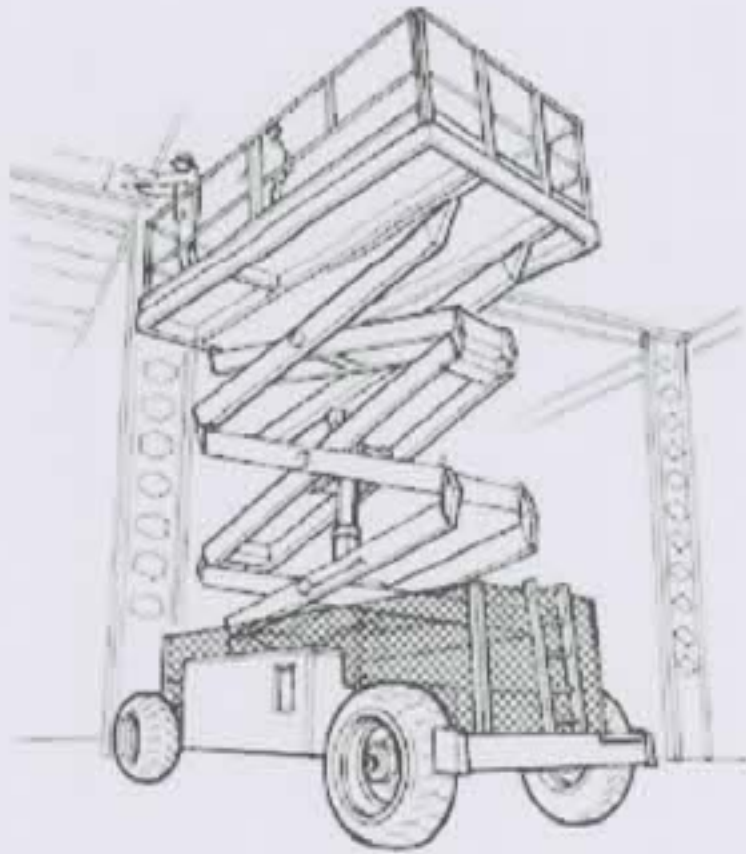


Figure 8.13 Mobile elevating work platform – one type of a wide range.

platforms are scissor lifts and boom type hydraulic platforms (cherry pickers). The following factors must be considered when using mobile elevated work platforms:

- ☐ the mobile elevated work platform must only be operated by trained and competent persons and safety harnesses should be available
- ☐ it should be inspected before use
- ☐ the working platform should not be overloaded
- ☐ it must never be moved in the elevated position
- ☐ it must be operated on level and stable ground with consideration being given for the stability and loading of floors
- ☐ the tyres must be properly inflated and the wheels normally immobilized when the platform is in use
- ☐ outriggers, if fitted, should be fully extended and locked in position
- ☐ warning signs should be displayed and barriers erected to avoid collisions
- ☐ due care must be exercised with overhead power supplies and obstructions and adverse weather conditions
- ☐ it should be maintained regularly and procedures should be in place in the event of machine failure
- ☐ drivers of MEWPs must be instructed in emergency procedures particularly to cover instances of power failure.

When working on a MEWP, there is a danger that the operator may become trapped against an overhead or

adjacent object and preventing him from releasing the controls. There have also been accidents caused when a MEWP is reversed into areas where there is poor pedestrian segregation and the driver has limited visibility. During any manoeuvring operation, a dedicated banksman should be used.

Other techniques

Sometimes it is necessary to work from a man-riding skip. It is important that the skip is of sound construction and made from good quality material which has adequate strength. It should be at least 1 m deep and have means to prevent tipping, spinning and persons from falling out. The skip should be properly maintained and be marked with the safe working loads. The crane and lifting tackle used to lift the skip must be regularly inspected and thoroughly examined with appropriate certification being obtained. Only competent persons should undertake this type of work and there must be proper communication with other interested parties, such as the building occupiers, the police and the local authority.

When maintenance operations take place on bridges or similar inaccessible structures, suspended access cradles or platforms may be used. The failure of access cradles has led to serious injuries and fatalities. There are many reasons for such incidents including:

- ☐ poor equipment selection procedures in that the cradle is not suitable for the purpose
- ☐ unsafe access to the cradle
- ☐ overloading of the cradle or non-uniform loading across the platform
- ☐ holes or cracks in the platform floor allowing material to fall below
- ☐ inadequate guard rails and toe boards
- ☐ the structure to which the cradle is attached is not capable of carrying the additional load
- ☐ insecure counterweights and/or braking system
- ☐ the failure of essential components of the cradle structure
- ☐ the failure of the winching and climbing devices
- ☐ poor erection, maintenance and dismantling by incompetent and untrained personnel
- ☐ inadequate emergency procedures.

The safest access to a cradle is at ground level. If access at height is unavoidable, then the cradle must be secured so that it is prevented from swinging away from the structure. It is also important that suitable anchorage points are available for safety harnesses and a secondary rope is fitted. Regular visual inspections should be made at least before each time that it is used and a weekly detailed inspection should be made by a competent person.



Figure 8.14 Mast cradle.

A boatswain's chair is often used for light work of a short term nature. Whenever it is used, the worker should be protected by a harness and lanyard in case of a possible fall. Abseiling or rope access techniques are also used for inspection work. Devices such as these should only be used when a properly constructed working platform is not practicable.

Equipment, such as cradles, boatswain's chairs and rope access techniques, must be erected either by or under the supervision of a competent person and the worker must be properly trained in the use of the technique. The main rope and safety access rope should always be attached to separate anchorage points. Any tools or equipment used must be secured to the worker to prevent them falling and possibly injuring people below. Whenever possible, the area beneath the work should be fenced off.

Fall arrest equipment

The three most common types of fall arrest equipment are safety harnesses, safety nets and air bags.

Safety harnesses should only be used alone when conventional protection, using guard rails, is no longer practicable. Such conditions occur when it is possible to fall 2 m or more from an open edge. It is important that the following points are considered when safety harnesses are to be used:

1. The length of fall only is reduced by a safety harness. The worker may still be injured due to the shock load applied to him when the fall is arrested. A free fall limit of about 2 m is maintained to reduce this shock loading. Lanyards are often fitted with shock absorbers to reduce the effect of the shock loading.
2. The worker must be attached to a secure anchorage point before they move into an unsafe position. The lanyard should always be attached above the worker, whenever possible.
3. Only specifically trained and competent workers should attach lanyards to anchorage points and work in safety harnesses. Those who wear safety harnesses must be able to undertake safety checks and adjust the harness before it is used.

Safety nets are widely used to arrest falls of people, tools and materials from height but competent installation is essential. The correct tensioning of the net is important and normally specialist companies are available to fit nets. The popularity of nets has grown since the Construction Regulations came into force in 2003 and the subsequent advocacy of their use by the HSE. Nets are used for roofing work and for some refurbishment work. Nets, however, have a limited application since they are not suitable for use in low-level construction where there is insufficient clearance below the net

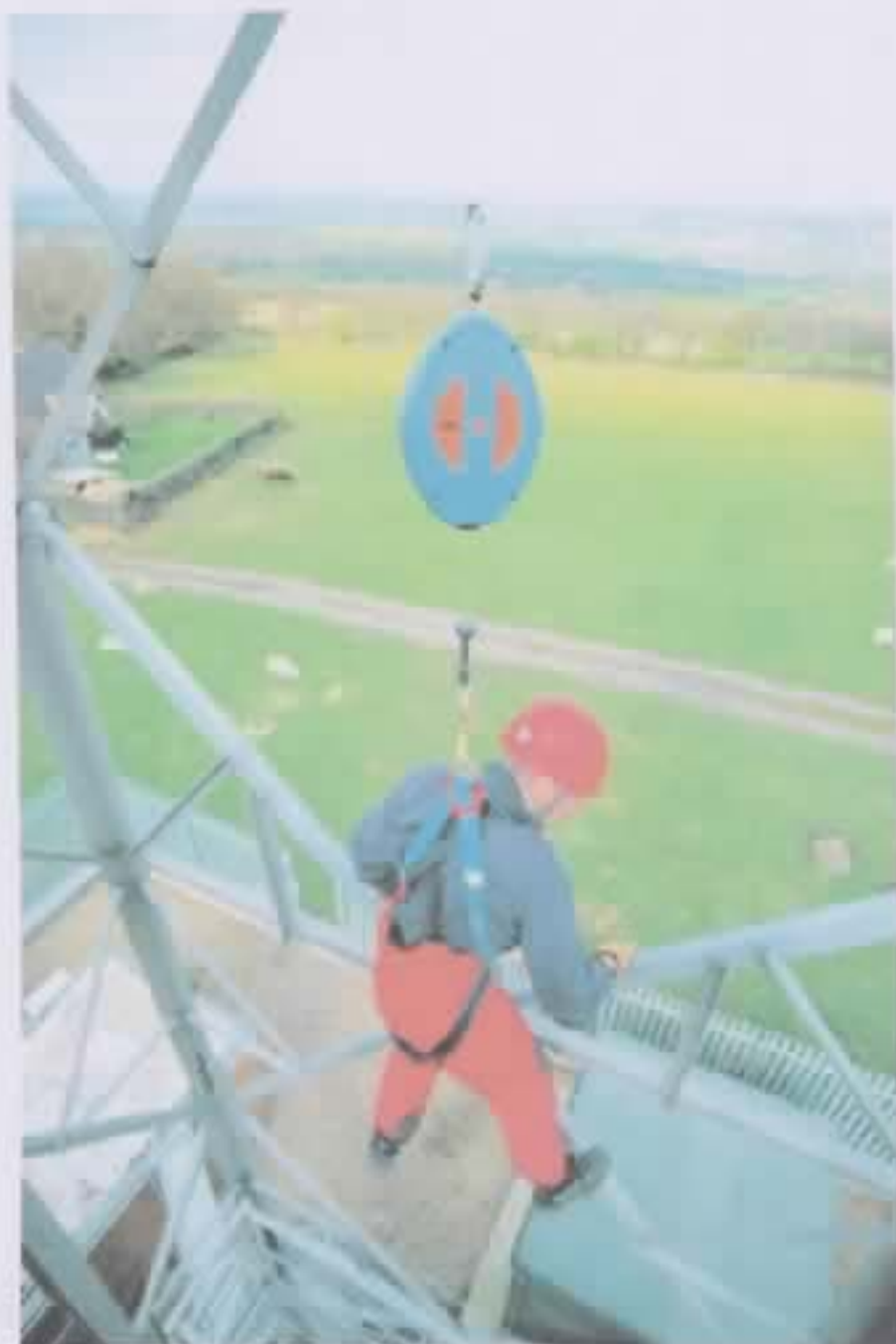
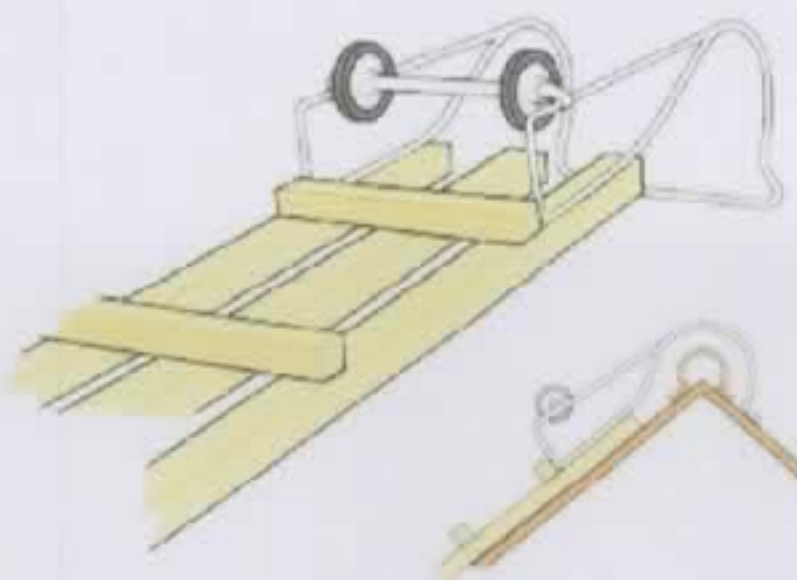


Figure 8.15 Fall arrest harness and device.

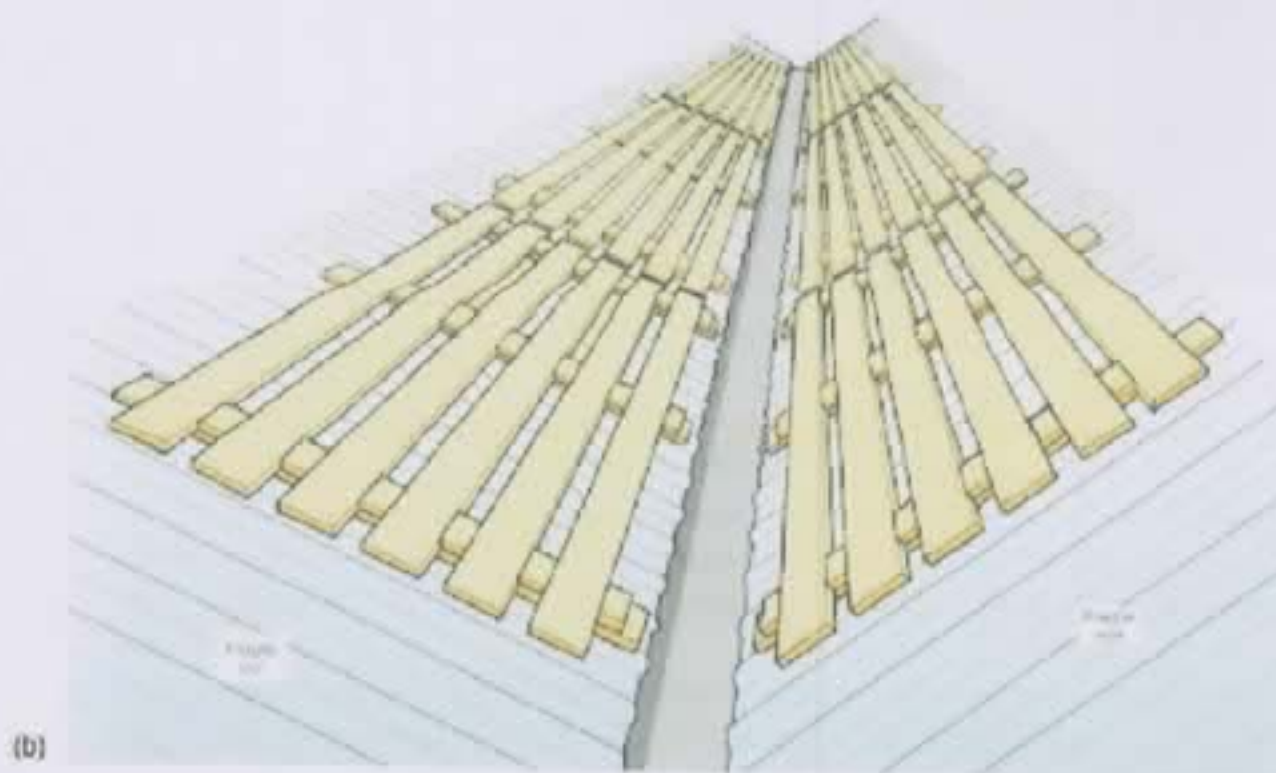
to allow it to deflect the required distance after impact. Nets should be positioned so that workers will not fall more than 2 m, in case they hit the ground or other obstructions.

Air bags are used when it is either not possible or practical to use safety nets. Therefore they are used extensively in domestic house building or when it is difficult to position anchorage points for safety harnesses. When air bags are used, it is important to ensure that the bags are of sufficient strength and the air pressure high enough to ensure that any falling person does not make contact with the ground. Only reputable suppliers should be employed for the provision of air bags.

After the completion of the construction project, cable-based fall arrest systems are normally suitable for on-going building maintenance work. They may also be used for the construction of complex roof structures,



(a)



(b)

Figure 8.16 (a) Roof ladder. The ridge iron should be large enough to be clear of the ridge tile. (b) Permanent protection installed at valley gutter (the protection should be supported by at least three rafters beneath the roof sheets).

such as parabolic or dome structures, where a safety net may be more than 2 m below the highest point of the structure.

8.4.3 Roof work

Before any roof work is started, a risk assessment should be undertaken. This should include an assessment of the structural integrity of the roof and the methods to be used to repair the roof and a COSHH assessment of any hazardous substances to be used. The risk assessment should be followed by the provision of a method statement by the sub-contractor. This should be examined to ensure a safe method of working is proposed and which does not conflict with other sub-contractors. Only competent and trained workers should be employed for roof work and any equipment used should be suitable for the job.

The precautions needed when repairing the flat roof of a building include the provision of safe access to the roof (using scaffolding and/or ladders), the use of crawling boards, the provision of safety harnesses and edge protection, using toe boards, to prevent the fall of persons or materials. Roof battens should be checked to ensure they provide safe hand and foot holds. If they do not then crawling ladders or boards will be used. Crawling boards should always be provided when work on fragile roofing materials cannot be avoided.

The perimeter of the roof requires a two-rail scaffold guard rail fitting together with a toe board where the gutter would not prevent materials slipping from the roof. Protection of the leading edge must also be provided. Where this is also used as a working platform, it is recommended that the 'Youngman' trolley system or similar fitted with guard rails is used. Suitable safety nets may also need to be provided to prevent people and materials from falling.

Access to the area immediately below the work should be restricted using suitable barriers, netting, safety signs and safety helmets. There should also be no danger to employees from fragile roof lights, voids, overhead obstructions and services.

All walkways for the passage of people and materials should be a minimum of 600 mm wide and protected with double guard rails and toe boards on the exposed edge. Where staging is used care should be taken to ensure that the ends are supported and not cantilevered over the steelwork. Where a gutter is used as a walkway either along the perimeter or along a valley of the roof the need for similar protection still applies. Walkways should be kept clear of materials, tools and debris as far as possible.

Consideration should also be given to the way in which materials are transported to and from the working

area, involving lifts, hoists, manual handling and/or using chutes and covered waste skips. When they are to be used on the roof, such materials should be secured in demarcated areas. Materials stacked on the roof should be securely tied down when not required to avoid the possibility of being blown down from the roof. Where materials are stacked on purlins on the open part of the roof, a suitable form of walkway as described above should be provided around each pack and also for the working area.

Good housekeeping procedures are essential for safe roof work.

All openings left in the roof for lights, vents or similar should either be protected with guard rails or have some form of temporary covers fitted.

Where an inclined hoist is used to lift materials onto the roof this should have protective barriers around the base, an overhead mesh guard fitted to protect the operator and around the top a suitable walkway with guard rails to provide a safe means of off-loading the materials onto the roof.

Ladders securely tied should be situated as close as possible to the working area on the roof and extend 1 m above the roof level. Where the height to the roof exceeds 9 m, it is normally a requirement that a scaffold tower with ladders be provided such that the height between intermediate platforms does not exceed 9 m.

If asbestos is present, then special precautions will be required including the provision of overalls and respiratory protection, the need for damping down and the avoidance of breaking asbestos sheets. Safe disposal of waste asbestos is particularly important. Consideration should also be given to the wearing of a safety harness where other forms of protection are not adequate. This applies for example when fixing the gutter or the scaffold guard rail at the start of work. Safety helmets should be worn by all persons in accordance with the Construction Head Protection Regulations whilst work is in progress overhead and/or there is a risk of head injury.

During the construction of large industrial roofs, a purlin trolley system is usually used together with a safety harness. Such systems have a double handrail on the leading edge positioned on the opposite side to the working side. As the working side is open to the roof, a safety harness is needed to protect against falls from this edge. A second line of defence has recently been designed into these trolleys by the attachment of a horizontal barrier beneath it.

Other issues include the use of netting and protection against adverse weather conditions, by issuing suitable clothing (and possibly sun cream) and limiting exposure. The sensible positioning of bitumen boilers and gas cylinders will also reduce the risk of fire and

electrical risks will be reduced by the use of residual current devices or reduced low voltage for portable electrical equipment.

One final aspect of roof work which must be considered is work that takes place immediately below the roof in the attic or roof space, where access may be gained using mobile elevated work platforms and 'Youngman' staging platforms. An example of this is the painting of roof girders in a warehouse. As with similar work, continuous liaison with the client is essential during the planning stage and after the work begins. At the planning stage, any hazards existing in the roof space must be assessed including any lighting, heating and electrical and other services that may be present. Often other work activities may be taking place in the warehouse while the painting work is in progress. It is essential, therefore, that there is agreement with the client on the sequence of work, working periods and working areas so that there is no accidental contact between painters and high reach pallet fork lift trucks. The painters will need to be briefed on the operational methods and routes used by the pallet trucks. Similarly, the warehouse workers will require information on the materials and systems of work used by the painters. Other related aspects of the work, such as safe use of the mobile and stationary platforms, the use of harnesses and debris netting, storage arrangements for paints and solvents and the provision of respiratory protective equipment have been dealt with elsewhere.

Three examples of safe systems of roof work are given in Appendix 8.4.

8.4.4 Protection of other persons

Members of the public must be protected at all stages of the construction project and at all times of day and night. Scaffold should be made very visible by painting standards red and white stripes, masking couplers, warning signs, good illumination at night and the provision of barriers and other means of safe passage. Safety precautions, such as the boarding of ladders, should be taken during periods when the site is unoccupied. Such protection should be provided to protect anyone including trespassers, such as children.

When construction work is being undertaken on occupied premises, such as a retail store or an office block, certain important health and safety matters would need to be discussed and agreed with store or office management. These matters would include the location and isolation of the construction working areas, supervision of the work, procedures for the evacuation of the premises in the event of an emergency, the use of welfare and canteen facilities, the protection of the occupier's employees and procedures for handing back

areas of the store when the work has been completed. The construction contractor must provide protected access into the building, ensuring that the area to be occupied by him is provided with adequate fences or barriers and that all waste materials are safely removed. Other issues which need to be addressed include the selection of equipment (preferably using reduced voltage electrical equipment) and personal protective equipment and the erection of appropriate warning signs. Other precautions and controls include the provision of:

- safe access for workers and materials
- a safe working platform
- clean-up procedures and good housekeeping
- full arrest equipment
- protection for all premises' occupier's employees
- a permit system for hot work where this was necessary
- fire extinguishers and procedures for emergency evacuation of the building.

Finally some or all of existing services may need to be isolated at certain times, and these times should be agreed with the occupier and adequate levels of lighting must be provided, where necessary, to carry out the work safely.

8.4.5 Inspections and maintenance

Inspection

Equipment for work at height needs regular inspection to ensure that it is fit for use. A marking system is probably required to show when the next inspection is due. Formal inspections should not be a substitute for any pre-use checks or routine maintenance. Inspection does not necessarily cover the checks that are made during maintenance although there may be some common features. Inspections need to be recorded but checks do not.

Scaffolds must be inspected on a regular basis by a competent person. These inspections should take place before the scaffold is used, after any alteration is made or after adverse weather conditions may have weakened it. In any event an inspection should take place every seven days and any faults rectified.

Under the WAHR, weekly inspections are still required for scaffolding, as previously required by the Construction (Health, Safety and Welfare) Regulations where a person could fall 2 m or more. The reporting requirements for inspection are set out in Schedule 7 to the regulations as follows:

- the name and address of the person for whom the inspection was carried out

- the location of the work equipment inspected
- a description of the work equipment inspected
- the date and time of the inspection
- details of any matter identified that could give rise to a risk to the health or safety of any person
- details of any action taken as a result of any matter identified in previous paragraph
- details of any further action considered necessary
- the name and position of the person making the report.

Whilst there is no longer a specific statutory form to be completed, a record of the inspection should be made on the recommended form for inspections shown in Appendix 3.1, with a list of typical scaffolding faults in Appendix 8.2 and a possible checklist, which could be used, is given in Appendix 8.3.

Most equipment used while working at height will need regular inspection and, in some cases, a thorough examination. Such equipment includes hoists and mobile elevated work platforms, grit-blasting and water-jetting equipment and electrical appliances.

Maintenance

Inspections and even thorough examinations are not substitutes for properly maintaining equipment. The information gained in the maintenance work, inspections and thorough technical examinations should inform one another. A maintenance log should be kept and be up to date. The whole maintenance system will require proper management systems. The maintenance frequency will depend on the equipment, the conditions in which it is used and the manufacturer's instructions.

Practice NEBOSH questions for Chapter 8

- Outline the precautions to be taken when carrying out repairs to the flat roof of a building.
- Outline the precautions that might be taken in order to reduce the risk of injury when using stepladders.
- Identify measures that should be adopted in order to protect against the dangers of people and/or materials falling from a mobile tower scaffold.
- Grit-blasting and concrete repairs are to be carried out on the external facade of a five-storey building. Access will be by an independent tied scaffold, which will take up the entire width of the adjoining pavement.

Outline the possible hazards associated with this work AND the precautions that should be taken.
- The exterior paintwork of a row of shops in a busy high street is due to be re-painted. Identify the hazards associated with the work and outline the corresponding precautions to be taken.
- As part of the refurbishment of a three-storey building within a busy city centre, a section of stone facade is to be cleaned using a specialist cleaning contractor. The work will be undertaken using high pressure water-jetting.

Describe a safe system and method of work to be adopted by the contractor to ensure the safety of:

 - the water-jetting operatives
 - the workers carrying out the internal refurbishment work
 - the public and users of the highway.
- The waterproof membrane on the flat roof of a two-storey, high security office block is about to be replaced. Access for personnel not engaged in the roofing work is prohibited but an access route from inside the building is available. The only external access route for materials is via a lightweight, inclined builders' hoist, which must be erected and dismantled daily.

Describe the procedures necessary to ensure:

 - the safety of the occupiers of the building
 - the safety of the operatives whilst working on the roof
 - the safe erection and dismantling, and use, of the inclined hoist.
- Minor repairs to the external fabric of a multi-storey office block are to be carried out. The chosen method of access to upper floors is by a man-riding skip suspended from the hook of a mobile crane.
 - Outline the specific requirements relating to the safe use of the skip.
 - Describe other measures that should be in place to ensure the safety of all persons affected by the work.
- A gable end wall of a three-storey building adjacent to a canal needs repointing. An independent tied scaffold will be used for access. The scaffold will obstruct the towpath and will restrict the width of the canal.

Outline the health and safety issues associated with this work and how they might be addressed.
- Sheet piles are to be driven into a river bed and the water contained by the piles pumped out before

construction work starts. Identify the health hazards that may be encountered by workers involved in this operation and explain how such hazards may be controlled.

11. Describe the specific measures that may be necessary to ensure safety when work is to be carried out from a scaffold that overhangs a fast-flowing river.
12. Outline the main points to be considered when selecting and inspecting a wooden pole ladder prior to use.
13. Outline the measures to be taken to prevent falls associated with stairwells and other holes in floors during the construction of a multi-storey building.
14. Scaffolding has been erected to the outside of a block of high-rise flats in order to undertake window replacement and repairs to external cladding.
Outline the factors that could affect the stability of the scaffold.
15. Explain the issues that would need to be addressed if work is to be carried out safely from a ladder.
16. Outline a hierarchy of measures to be considered when a construction worker is likely to fall while working at height.
17. List EIGHT items to be examined when inspecting a putlog scaffold.
18. A four-storey building on the main street of a busy town centre requires external repairs to its fabric. The scaffold to be erected for the work will take up the entire width of the pavement.
 - (i) Sketch the front and end elevation of a suitable scaffold for the work, clearly labelling the key safety features.
 - (ii) Identify, by means of a sketch where appropriate, the measures that will be needed to ensure safe passage for pedestrians and traffic.
 - (iii) Outline the procedures to be adopted in order to ensure the safety of employees using the scaffold as a working platform.
19. A row of terraced houses is to be built alongside an existing public road that runs up a slope.
 - (i) Sketch the front and end elevation of suitable scaffold for this work, clearly identifying the key safety features.
 - (ii) Outline the procedures and practices required to ensure the safe erection and use of the scaffold in (i).
20. A mobile tower scaffold is to be used in the re-pointing of external brickwork on the gable end wall of a building, which is 10 m high at its highest point.
 - (i) Explain the possible dangers associated with this operation.
 - (ii) Outline, with the aid of diagrams, the precautions to be taken to minimize the risks.
21. Extensive repairs are to be carried out to the flat roof of a two-storey office building, which is to remain in use during the work. A small platform hoist is to be used for lifting materials to the roof. There is no access to the roof from inside the building.
Outline the procedures and protective measures needed to ensure:
 - (i) the safety of occupiers of the building
 - (ii) the safety of operatives whilst work is carried out on the roof
 - (iii) the safe erection and use of the hoist and other access equipment.
22. A partial upgrade of an existing ventilation system is to be undertaken in a large supermarket. The work is to be carried out on a progressive overnight basis over 10 nights, when the only other activities in the store are shelf replenishment and cleaning. The majority of the work will involve access above a suspended ceiling at a height of 5 m above the ground but plant situated on the roof of the store is also to be replaced.
Describe the main features that must be covered to ensure the health and safety of contractors and others on site during the work.
23. Identify, by means of a labelled sketch, the main requirements for a loading bay on a scaffold in order that materials for bricklayers and roofers can be safely placed on the bay by a fork-lift truck.
24. A painting contractor has been contracted to paint roof girders in a racked storage warehouse. Painters will gain access to the roof area from both mobile elevating work platforms and 'Young-man' staging platforms. High-reach pallet trucks will continue to be used in the warehouse during the work.
Explain how the work should be planned and undertaken in order to ensure the health and safety of both painters and warehouse staff.
25. A wooden pole ladder is to be used as a means of access to a scaffold. Outline the features of the ladder that should be considered, and the particular items that should be inspected, in order to ensure its suitability for the job.

Appendix 8.1 – Inspection report form

Report of results of every inspection made in pursuance of Construction (Health, Safety and Welfare) Regulations 1996 or the Work at Height Regulations 2005

1. Name and address of person for whom inspection was carried out.

2. Site address

3. Date and time of inspection.

4. Location and description of workplace (including any plant, equipment or materials) inspected.

5. Matters which give rise to any health and safety risks.

6. Can work be carried out safely?

Y / N

7. If not, name of person informed.

8. Details of any other action taken as a result of matters identified in 5 above.

9. Details of any further action considered necessary.

10. Name and position of person making the report.

11. Date report handed over.

Inspection reports: Notes

Place of work requiring inspection	Timing and frequency of inspection					
	Before being used for the first time.	After substantial addition, dismantling or alteration.	After any event likely to have affected its strength or stability.	At regular intervals not exceeding 7 days.	Before work at the start of every shift.	After accidental fall of rock, earth or any material.
Any working platform or part thereof or any personal suspension equipment.	@	@	@	@		

NOTES

General

1. The inspection report should be completed before the end of the relevant working period.
2. The person who prepares the report should, within 24 hours, provide either the report or a copy to the person on whose behalf the inspection was carried out.
3. The report should be kept on site until work is complete. It should then be retained for three months at an office of the person for whom the inspection was carried out.

Working platforms only

1. An inspection is only required where a person is liable to fall more than 2 metres from a place of work.
2. Any employer or any other person who controls the activities of persons using a scaffold shall ensure that it is stable and of sound construction and that the relevant safeguards are in place before his employees or persons under his control first use the scaffold.
3. No report is required following the inspection of any mobile tower scaffold which remains in the same place for less than 7 days.
4. Where an inspection of a working platform or part thereof or any personal suspension equipment is carried out:
 - i. before it is taken into use for the first time; or
 - ii. after any substantial addition, dismantling or other alteration; not more than one report is required for any 24 hour period.

Appendix 8.2 – Checklist of typical scaffolding faults

Footings	Standards	Ledgers	Bracing	Putlogs and transoms	Couplings	Bridles	Ties	Boarding	Guard rails and toe boards	Ladders
Soft and uneven	Not plumb	Not level	Some missing	Wrongly spaced	Wrong fitting	Wrong spacing	Some missing	Bad boards	Wrong height	Damaged
No base plates	Jointed at same height	Joints in same bay	Loose	Loose	Loose	Wrong couplings	Loose	Trap boards	Loose	Insufficient length
No sole plates	Wrong spacing	Loose	Wrong fittings	Wrongly supported	Damaged	No check couplers	Not enough	Incomplete	Some missing	Not tied
Undermined	Damaged	Damaged			No check couplers			Insufficient supports		

Appendix 8.3 – A checklist for a safety inspection of a scaffold

- Is the scaffold on a firm foundation?
- Are standards (uprights) resting on suitable base plates and (where the scaffolding is not on hard standing) timber sole boards? Bricks, blocks and other building materials should not be used as packing.
- The scaffold should not be undermined by excavations close to its supports.
- Are the uprights vertical and the horizontals horizontal?
- Are the uprights close enough together and is the spacing consistent? (Note: the permissible width of bays will vary with the operations for which the scaffolding is intended. The greater the loading, the closer the uprights will need to be.)
- Are load-bearing couplers used where appropriate?
- Are working platforms properly supported (each board resting on at least three supports – no board should overhang the last support by a distance greater than three times its thickness), fully boarded out (no gaps for people or materials to fall through) and wide enough to allow safe access and the safe movement of materials?
- Has the scaffold been erected by a competent person? Are inspection records in order?
- Are all components in good condition?
- Are guard rails and toe boards fitted to all working platforms to prevent people and materials falling?
- Is additional protection such as wire mesh brick guards used where appropriate?
- Is the scaffold loaded with excessive quantities of materials?
- Are materials safely stacked? (Loads such as piles of bricks should be in line with the standards.)
- Is the scaffold structure adequately stiffened by the use of diagonal bracing?
- Is the scaffold adequately tied to a structure, or suitably buttressed, to prevent collapse?
- Are incomplete sections of the scaffold marked with suitable warning notices?
- Where there is sheeting or some other feature which will increase the windage of the scaffold, has this been allowed for in the design?
- Is there safe access to all the scaffold's working platforms?
- Are there safe arrangements for the raising and lowering of materials?
- Are there adequate barriers etc. to keep people who are not involved in the work clear of danger?

Appendix 8.4 – Examples of safe systems of work used in roof work

1. Slating work

1. Materials will be delivered as close as possible to the roofing operation.
2. Materials will be loaded to the scaffold by a mechanical conveyor or ladder.
3. Felt and battens will be laid from the eaves to the ridge in accordance with the specifications.
4. All slating will progress from right to left across the roof and, on completion, the ridge tiles and fixings will be similarly loaded and laid.
5. All waste materials and debris will be cleared from the site daily.
6. The site will be left in a tidy condition on completion.

2. Lead work on roofs

1. The Control of Lead at Work Regulations 2002 and associated Approved Code of Practice will be followed closely during all work involving lead.
2. Appropriate manual handling procedures will be adopted.
3. Work will be confined to clean solid metallic lead and always carried out in the open air.
4. In areas where lead work is being carried out, no food or drink will be consumed and no smoking allowed.
5. Adequate washing facilities will be provided on site. Nail brushes will be provided.
6. All employees working with lead will wear overalls which must remain on site at all times (except when being laundered at a suitably equipped facility).

3. Working on a fragile roof (with a non-asbestos roofing material)

A risk assessment will be provided for each job.

1. The area of roof in question which is to be repaired or renewed will be assessed from the scaffold unless a specific scaffold extension or duckboards have been provided and in place on the area in question.
2. Fixing bolts will be removed one at a time either manually or with the use of 110 volt portable equipment. No smashing or breaking of sections of sheeting is permitted unless prior authorization has been obtained from the operations manager.
3. Old sheets and material are to be placed into the skip provided via the installed rubbish chute. In the event of larger sections which will not go into the chute, being removed they will be lowered by hoist or pulley wheel to ground level where they will be received by a colleague.
4. In the event of repair of a fragile roof by walking through the valley gutters the following precautions must be taken:
 - (a) The valley will be cleared of debris and water, i.e. leaves and soil etc. Any areas where the valley is showing dips or deflection should be reported immediately, unless they were briefed at the start of the work, for a revision of the work method.
 - (b) Both elevations of the roof either side of the gutter will be provided with either a continuous 'safe site' supergrip platform to reduce the possibility of slipping and falling through fragile sheeting or a series of good condition pallets will be used to serve the same purpose.
5. To carry out leak spotting, a safety harness secured to either an internal roof purlin by removing a ridge cap or other secure fixing near the ridge.
6. To provide a pathway on fragile sheets a series of duckboards should be used with specially adapted hook assembly on inclined roof areas.
7. Under no circumstances is anyone allowed to walk on sheets, bolts, or any other bare roof coverings to gain access to an area requiring attention.
8. If workers are unable to use equipment provided they must report to the operations manager and withdraw from the area until receiving further instructions.

END OF TRAINING



